

ROLANDS BARKANS

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EDUCATION

Minerva University (1% acceptance rate) – San Francisco, CA

Aug 2023 – May 2027

B.S. in Data Science and Statistics; Earth and Environmental Systems | GPA: 3.74/4.0

- Relevant Coursework: Machine Learning, Bayesian and Causal Inference, Algorithms and Data Structures, Linear Algebra, Atmospheric Chemistry, Remote Sensing, Geomorphology, Geochemistry

EXPERIENCE

Data Engineering Intern – Tradewater

Jun 2026 – Present

Chicago, IL

- Designed and deployed automated ETL pipelines integrating 7 external APIs and CRM systems into SQL Server, with focus on reducing execution time and associated cloud compute costs
- Structured data warehouse across three layers with dedicated schemas for current state, historical snapshots, and period-over-period changes to enable consistent and auditable reporting across organization
- Developed 5+ Power BI dashboards for internal stakeholders, gathered requirements through direct collaboration, and iterated on reports based on feedback to support sales pipeline and operational decisions

Data Analytics Intern – Sun Spark Group

Sep 2024 – Present

Taoyuan City, Taiwan

- Architected GCP data pipeline with Cloud Run Functions to ingest POS sales data into BigQuery; designed Looker dashboards to visualize real-time KPIs for operational decisions
- Trained ARIMA, Random Forest, and XGBoost forecasting models with 96% accuracy for ingredient demand forecasting, cutting food waste by 12%
- Optimized delivery logistics with Google OR-Tools route optimizer to reduce total delivery time by 34% and automate 90% of manual planning

Software Engineering Intern – Children Helping Science

Sep 2024 – Apr 2026

San Francisco, CA

- Automated end-to-end data sync across 50+ studies daily with Selenium and Google Apps Script to reduce manual data entry and save 20 hours per week
- Wrote internal documentation and tutorials adopted by 10+ contributors to decrease onboarding time by 50% and reduce workflow errors by 30%
- Maintained Python scripts for continuous dataset cleaning, archiving, and updates across active research studies

Research Assistant – Harvard University

Sep 2023 – Aug 2024

San Francisco, CA

- Cleaned and validated 350+ child study records to produce structured datasets ready for statistical analysis
- Proposed study design linking student empathy to music habits, operationalized through Perth Empathy Scale and EMMA database
- Synthesized scientific literature into concise reports to guide experimental design and variable selection

PROJECTS

East Africa Drought Prediction

- Trained CNN-LSTM on ERA5-Land reanalysis data (1980–2025) to predict monthly drought severity at pixel level across East Africa with 86% accuracy; automated plain-language summaries of monthly maps via OpenAI API

Taxi Trip Duration Prediction

- Built LightGBM quantile regression model on NYC yellow taxi data to predict trip duration as confidence intervals with median MAE of 0.8 minutes and 79.8% of trips inside predicted range

Soot-Driven Glacier Loss Simulation

- Built Monte Carlo cellular automata model simulating black carbon effects on glacier melt over 40 years on 100×100 spatial grid; found 80% emissions reduction extends glacier survival by 3.6 years and cuts peak meltwater discharge by 17%

Patagonia Glacier Albedo Modeling

- Modeled relationship between surface reflectivity and ice elevation change across Southern Patagonian Icefield (2000–2015) using NASA MODIS satellite data and Bayesian regression in PyMC, comparing five model specifications

SKILLS

- **Programming:** Python (NumPy, Pandas, xarray, Scikit-Learn, PyTorch, PyMC, Matplotlib, Seaborn) · SQL · R
- **Machine Learning:** feature engineering · gradient boosting · deep learning (CNN, U-Net, LSTM, ConvLSTM) · PCA · EDA · Bayesian inference · time-series forecasting · quantile regression
- **Data Engineering:** ETL pipeline design · API integration · medallion architecture · stored procedures · data quality checks
- **Cloud & Infrastructure:** Azure Functions · Azure SQL · GCP · BigQuery · Cloud Run · AWS
- **Tools:** Power BI · Looker · Git · Google Earth Engine · GeoPandas · OR-Tools · Claude · ChatGPT · Gemini · Perplexity